

Robotic animals are already here

Artificial intelligence. A joint project by University of Essex and Strathclyde, the Tyndall National Institute and Thales Safare (TSFR), this 152cm (five feet) long robot swims like real fish and is fitted with sensors that collect samples and determine the pollution level in water bodies. This robo-fish can detect and avoid obstacles, communicate with each other, send data to shore station up to 1km away and return on its own to base when the 8-hour long battery runs low.

Ready to run. Boston Dynamics has developed WildCat robot which runs at a speed of over 29mph (faster than Usain Bolt) and has set a new land-speed record for legged robots. The design of Wildcat and its predecessor Cheetah are inspired by locomotion of animals: an articulated back that flexes back and forth on each step, thereby increasing stride and running speed.

Snakes. A 1.2cm (half inch) in diameter snake-robot named MiRoR (Miniaturised Robotic systems for holistic in-situ Repair and maintenance) works in restrained and hazardous environments to allow engine experts to remotely and quickly find faults in large, complex machines like aircraft. Apart from UV laser to perform welding and grinder to sand down and smoothen damaged parts, this robot is also equipped with self-positioning, reasoning, planning and adaptation capabilities. MiRoR is a joint project of Rolls-Royce, University of Nottingham, Fraunhofer IPA, IK4 Tekniker, ETH Zurich, Acciona Infrastructure and OMV Petrom.



some biological characteristics of the animal it is pretending to be.

For example, PARO has five kinds of sensors: tactile, light, audition, temperature and posture. It processes data from these sensors to collect information with which it can perceive people and its environment. With the light sensor, the robot can recognise light and its intensity. It feels being stroked and beaten by tactile sensors, or being held by posture sensor. It can also recognise the direction of voice and words such as its name, greetings and praise with its audio sensor.

Researchers at National Sun Yat-sen University have developed a framework that employs a neural network-based approach to construct behavior primitives and arbitrators for robots. This neural network-based architecture could allow these researchers to build a robot that learns or evolves its behavior based on how the human interacts with it, or based on the data it gets from its sensors. They believe that it will be far easier to implement through this architecture, compared to traditional solutions.

PhD student Mriganka Biswas says in an article in *ScienceDaily*, "Cognitive biases make humans what they are, fashioning characteristics and personality, complete with errors and imperfections. Therefore introducing cogni-

tive biases in a robot's characteristics makes the robot imperfect by nature, but also more human-like. Based on human interactions and relationships, we will introduce characteristics and personalities to the robot. If we can explain how human-to-human long-term relationships begin and develop, it would be easier to plan the human-robot relationship."

Step four: takeover their vehicles

OEMs have taken autonomous cars seriously, and Land Rover has taken its first step towards self-driving, cloud-connected, augmented-reality based vehicles. Discovery Vision is Land

Rover's all new SUV concept expected to hit roads in a few years. BMW and Audi are coming out with their own versions, although none have reached the levels of Transformer robots yet.

NVIDIA's Tegra K1 mobile processor is currently the processor of choice for Audi, as the German automaker tips its toes into autonomous vehicles. The K1 processor is powered by a 192-core graphics processing unit (GPU) which is the heart of Audi's entire automotive infotainment system. This gives an idea of the level of computing required for an automotive robot to function properly.

Of course, you need some very sophisticated software to ensure that your car doesn't end up jumping over a bridge with you inside. Bosch is developing an Automatic Park Assist technology that will be out next year, which allows a car to be parked remotely using just a smartphone app. Another technology, Traffic Jam Assistant, will step in when the vehicle is moving at low speeds. Google is already trying to make fully automated driving a reality. Bosch's autonomous technology gathers data using an array of sensors, including radar and video cameras, as well as a roof-mounted laser scanner (LIDAR) that generates a detailed 3D map of the environment.

While most of these technologies can be connected to those used in other autonomous robots, safety is critical for the automotive robots, so these have lot more policies and regulations to

Robots that can understand humans

Feeling low? Take Pepper. Aldebaran Robotics and Softbank have unveiled Pepper, a cute affectionate robot that studies your expressions, gestures and speech tone to read your mood, cracks jokes and sings songs for you when you are feeling low. It comes with a multitude of sensors, 3D cameras, four microphones and advanced voice-recognition hardware and software. Pepper could serve as an emotional companion and an understanding friend.

The mind reader. Remember Rosie, the robot maid from the cartoon series *The Jetsons*? You do not have to wait too long for a real-life robot like Rosie that can tend to your needs. Developed by Personal Robotics Lab at Cornell University, this clever robot can be trained to be a wait staff and predict your actions with surprising accuracy. It performs functions like refilling your tea cup or opening the door without you needing to ask for it.

Prison-guard robot. Robo-guard is a South Korean correction service robot that observes inmates in a prison using embedded 3D cameras, detects changes in behaviour of inmates with the help of special software, and raises an alarm if anything suspicious is detected. It is an autonomous robot that can also be controlled by iPad.

Family robots. Brainchild of MIT professor Dr Cynthia Breazler, JIBO project has received incredible welcome and crowd-funding in Indiegogo community. It is a small robot that can see, hear, speak, recognise emotions, adapt to situations and possibly perform many other activities. Since it is based on open source platform, the skills of JIBO would only improve with time.